



SOP-20-880 Series Smart Operating Panel User Guide



Industrial
Automation



Intelligent
Elevator



New Energy
Vehicle



Industrial
Robot



Rail
Transit



Data code 19010611A02

Preface

Thank you for purchasing the SOP-20-880 series smart operating panel developed by Inovance.

The SOP-20-880 smart operating panel is a new generation of commissioning assistant that supports HE and MD880 series products. SOP-20-880 features a wide power supply range and LCD display. It supports multibus and applies to the single-drive/multi-drive systems. The operating panel provides functions such as parameter setting, state monitoring, parameter copy, fault analysis, program download, and USB relay/disk modes.

This user guide describes correct use of the SOP-20-880 series smart operating panel, including safety precautions, product information, installation and wiring, function and application, and maintenance. Read through the user guide before use. Contact Inovance if you have any question during use.



Precautions

- ◆ Drawings in the user guide are sometimes shown without covers or protective guards. Remember to install the covers or protective guards as specified first, and then perform operations in accordance with the instructions.
- ◆ Drawings in the user guide are shown for description only and may not match the product you purchased.
- ◆ The instructions are subject to change, without notice, due to product upgrade, specification modification as well as efforts to increase the accuracy and convenience of the user guide.
- ◆ Contact our agents or customer service center if you have any question during use.

Revision History

Date	Version	Description
June 2020	A00	First release
August 2021	A01	Made minor corrections.
June 2024	A02	Updated sections 2.5.2, 4.4, 4.5, 4.6, and 4.7.

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1 Safety Precautions

The following signs are used for safety information in this manual. The description of the safety signs is important and must be followed.



Note: Failure to comply may result in equipment damage or personal injury.

1.1 General Safety Precautions

This section describes precautions that must be observed during acceptance, storage, handling, installation, wiring, operation, inspection, and disposal.

1.2 Unpacking Inspection

Item	Description
Check whether the delivered product is consistent with your order.	The box contains the equipment ordered. Confirm the equipment model according to the nameplate.
Check whether the product is intact.	Check whether the product delivered is in good condition. If there is any part missing or broken, contact Inovance or your supplier immediately.

1.3 Precautions for Storage and Handling



NOTE

- ◆ Keep the equipment in a place that meets the following conditions:
 1. Away from direct sunlight.
 2. Ambient temperature does not exceed the limit.
 3. Relative humidity does not exceed the limit.
 4. Without dramatic temperature change or water condensation.
 5. Away from corrosive, flammable gases, dust, salty elements, and metal dust.
 6. Away from locations with water, oil or medicinal drips.
 7. Away from equipment that may generate vibration or shock.
- ◆ If the environment is harsh, it is recommended to use high-protection options for protection against water and dust.
- ◆ Do not lay the equipment one above another. Failure to comply may result in personal injury or equipment damage.

1.4 Installation Precautions



NOTE

- ◆ Do not install the equipment in environments with water or corrosive substance. If higher level of protection is needed, use options with higher degree of protection.
- ◆ Do not subject the equipment to combustibles. Failure to comply may result in electric shock or a fire.
- ◆ Install the product inside a cabinet with fire-proof and electrical protection. Failure to comply may result in a fire.
- ◆ Prevent intrusion of unwanted objects into the equipment. Failure to comply can result in faults and fire accidents caused by aging components.
- ◆ Do not exert strong impact on the equipment. Failure to comply may result in faults.

1.5 Wiring Precautions

Precautions

- ◆ SOP-20-880 can be connected to Inovance HE series, MD580, and MD880 series products through standard network cables.
- ◆ For use of SOP-20-880 with other products, consult with Inovance first. Failure to comply may result in equipment damage or fault.
- ◆ Do not use a crossover or damaged network cable. Failure to comply may result in damage or malfunction of SOP-20-880.
- ◆ Use the USB cable delivered by default or provided by Inovance to connect SOP-20-880 to the PC. For detailed wiring instructions, see the wiring section.
- ◆ Do not route the power cable and the USB cable through the same duct or bundle them together. Keep the power cable at least 30 cm away from the signal cable.
- ◆ Take proper shielding measures in the following scenarios to prevent equipment damage:
 - ① Locations with strong electric field or magnetic field
 - ② Locations with radioactive rays

1.6 Maintenance Precautions

Precautions

- ◆ The battery of SOP-20-880 can be replaced only by professionals after power-off.
- ◆ The SD card can be removed or replaced only by professionals after power-off.
- ◆ Do not use the gasoline, diluent, alcohol, acidic or alkaline detergent during cleaning to prevent enclosure discoloration or damage.
- ◆ Do not scratch the LCD display by sharp objects to prevent damage to the display.

1.6.1 Regular Inspection

Perform regular inspection according to the following table.

Type	Inspection cycle	Item
Routine check	Routine	Check the ambient temperature, humidity, dust, and foreign objects.
		Check whether unusual noise is generated.
		Check whether the USB connector cover and the battery cover are installed properly.
		Check whether the LCD display is normal.
		Check whether the key operates properly.
		Check whether unusual smell is generated.
Regular check	One year	Check whether the fastening parts get loose.
		Check whether the equipment is overheated.
		Check whether the USB connector gets damaged.
		Check whether the fastening parts of USB connector and RJ45 interface are loosened.
		Check whether the SD card is installed properly.
		Check whether the battery is installed properly.
Regular replacement	About 10 years	Replace the button battery.

1.6.2 Disposal Precautions

 NOTE
◆ When disposing of the products, observe any applicable regulations or laws on recycling and reuse of electronic products.

2 Product Information

The SOP-20-880 smart operating panel is a new generation of commissioning assistant developed by Inovance. It supports HE, MD580, and MD880 series products. SOP-20-880 has a wide power supply range and LCD display. It supports multiple buses and applies to the single-drive/multi-drive systems. It provides such functions as parameter setting, state monitoring, parameter copy, fault analysis, program download, USB relay mode, and USB disk mode.

2.1 Nameplate and Model Number

SOP-20-880

①②③

①	Product series SOP: Smart operating panel	③	Product type 880: 880 and HE series supported
②	Product type 20: Operating panel series		

MODEL: SOP-20-880

S/N: 01610168YG100099



Suzhou Inovance Technology Co., Ltd.

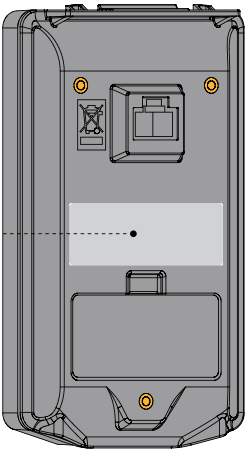


Figure 2-1 Nameplate description

2.2 Dimension Drawing

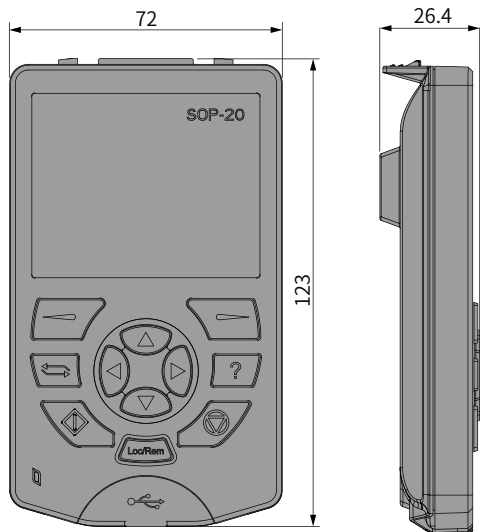


Figure 2-2 Dimension drawing of SOP-20-880 (unit: mm)

2.3 Components of SOP-20-880

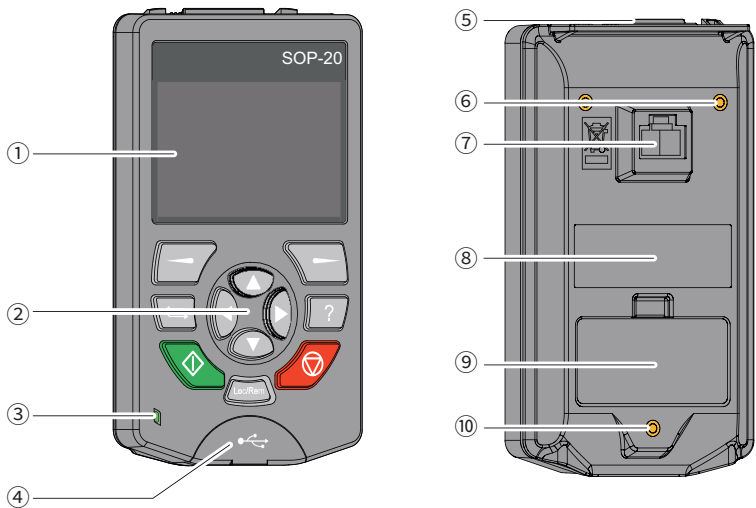


Figure 2-3 Components of SOP-20-880

No.	Name	Description
①	Display	-

No.	Name	Description
②	Key	-
③	Status indicator	-
④	USB port	Connect the SOP-20-880 to a PC with an USB cable for data exchange. After being connected to the PC, SOP-20-880 can work in relay mode as a PC tool (InoDriveShop) for data communication with the drive, or work in the USB disk storage mode to be used for SD card of SOP-20-880.
⑤	Snap-fit joint	The snap-fit joint on the stop is used to fix SOP-20-880 to the base.
⑥	Fixing screw	SOP-20-880 can be installed to the cabinet door by using the installation kit (IP20) and fixing screws. When the system complies with IP54, you may select the installation kit with higher IP rating (IP54).
⑦	RJ45 interface	Used to connect SOP-20-880 to the drive.
⑧	Nameplate	The nameplate displays the product model and serial number.
⑨	Battery cover	The battery and SD card slot are beneath the cover. The battery provides power to the real-time clock. The SD card stores factory parameters and real-time data of the drive to facilitate fault diagnosis and analysis. The SD card can also be used to store user program and update firmware or device.
⑩	Fixing screw	-

2.4 Introduction To the Interface

SOP-20-880 adopts the monocolour LCD with white backlight and a resolution of 240x160 pixels.

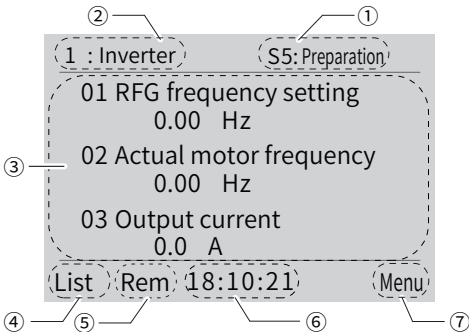


Figure 2-4 Interface example

Table 2-1 Interface description

No.	Display	Description
①	Device status and fault information	Displays the operating state of current device.
②	Current device information	1: Current device station number; Inverter: Current device name
③	Content	Real-time display of monitored information
④	Device list	Used to enter the device list of the current network.
⑤	Local/Remote	Local: SOP-20-880 has the control permission. Remote: SOP-20-880 has no control permission. Blank: This function is unavailable in the equipment or the equipment is connected improperly.
⑥	Clock	Displays current time.
⑦	Menu	Used to access the function menu options of current device.

2.5 Introduction to Keys

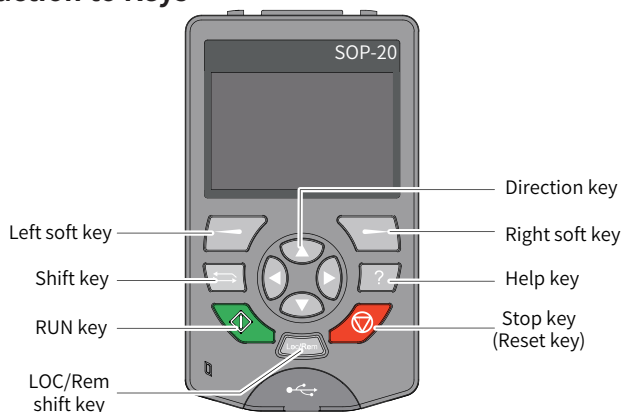


Figure 2-5 Position of the keys

2.5.1 Description of the keys

Table 2-2 Name and function

Key	Name	Function
	Left soft key	Used for exit or cancellation.
	Right soft key	Used for selection or confirmation.

Key	Name	Function
	Shift key	Used to enter the device list quickly. In some interfaces, you are not allowed to switch the devices. In this case, this key is invalid.
	Help key	Used to open the help page. The help page depends on the context, which means the contents of the help page is related to the menu or view.
	RUN key	Used to start the equipment in the local control mode.
	Stop key	Used to stop the equipment in the local control mode. When the device is in the fault state, use this key to reset the fault.
	Loc/Rem shift key	Used to switch between local and remote control.
	Direction keys	The up and down keys are used to select options in a displayed menu and list, scroll up and down a text page, and adjust a value (such as setting time, entering a password, or changing a parameter value). The left and right keys are used to move the cursor left and right.

2.5.2 Shortcut Keys

Table 2-3 Description of the shortcut keys

Shortcut key	Available in	Function
Left+Up	Any view	Used to adjust the backlight brightness.
Left+Down		
Right+Up	Any view	Used to adjust the contrast.
Right+Down		
Left+Help	Parameter list view	Used to add the parameter where the cursor is directed at to the shortcut interface.
Right+Help	Parameter list view	Used to add the parameter where the cursor is directed at to the boot screen.
Left+Help	Shortcut view	Used to delete the parameter from the shortcut commissioning interface.
Right+Help	Shortcut view	Used to delete all the parameters from the shortcut interface.
Left+Help	Boot screen view	Used to delete the parameter from the boot screen.
Right+Help	Boot screen view	Used to clear all parameters in the boot screen.

2.6 Indicator Status

The status indicator on the SOP-20-880 is used to indicate whether a fault or alarm exists.

Table 2-4 Indicator status

Symptom	Example	Meaning
Steady green		The equipment is running properly.
Flashing in green		Data is being transferred between the SOP-20-880 and PC with USB.
Steady red		Communication between the SOP-20-880 and the equipment is normal, but the equipment is faulty.
Flashing in red		Communication between the SOP-20-880 and the equipment has been interrupted.

2.7 General Specifications

No.	Type	Operation	Storage	Transportation
1	Operating temperature	-20°C to +55°C	-40°C to +70°C	-40°C to +70°C
2	Attitude	< 4000 m	-	-
3	Relative humidity	95%, without condensation		
4	Pollution degree	3C3 (IEC 60721-3-3, IEC 60721-3-2, and IEC 60721-1-3-1)		
5	Vibration	61800-5-1 ed 2 EN 60082-2-6 test Fc (1g)	-	2M3 class (according to EN 60082-2-6)
6	Shock	3M4 class (according to EN 60062-2-27)	-	2M2 class (according to EN 60082-2-27)
7	Free drop	IEC-60068-2-32 Drop height: 1 m	-	-
8	IP rating	When connect to drive equipment: IP54 (IP54 options used) Single unit: IP20		
9	Electromagnetic compatibility	IEC 61800-3		
10	Material	Disposal: The SOP-20-880 is recyclable. See related international and local laws and regulations for disposal and recycling.		

3 Installation and Wiring

The SOP-20-880 is suitable for multiple Inovance drives. This user guide takes M880 as an example to describe the installation and wiring of SOP-20-880. If you need to use SOP-20-880 for other drives, see the related user guides.

3.1 Installing SOP-20-880 to the cover of HCU

1) Connecting SOP-20-880

- ① Place the bottom of SOP-20-880 to the bottom ① of the slot on the HCU cover.
- ② Turn SOP-20-880 at the rotary point ① and push the upper part ② until you hear a click.

2. Taking out SOP-20-880

- ① Press and hold down the clip ② with the index finger and middle finger to release SOP-20-880.
- ② Pull the upper end of SOP-20-880 out of the HCU cover slot.

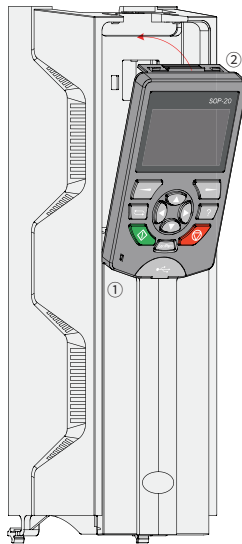


Figure 3-1 Installing SOP-20-880 to the HCU cover

3.2 Installing SOP-20-880 To the Cabinet Door

When the SOP-20-880 is installed to the cabinet door, secure the base of SOP-20-880 to the cabinet door with the four screws ③ shown in Figure 3-2. Then install SOP-20-880 to the base based on section [“3.1 Installing SOP-20-880 to the cover of HCU”](#) .

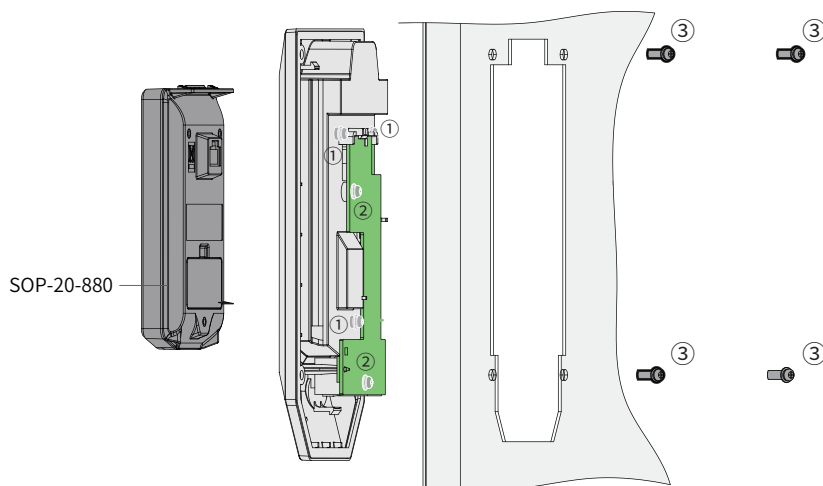


Figure 3-2 Installing base to the door plate

As shown in Figure 3-2, ① represents screws used to fix SOP-20-880, which must be used with IP54 protection. ② represents screws used to fix the PCB, with the grounding function provided. ③ represents screws used to fix the base to the cabinet door.

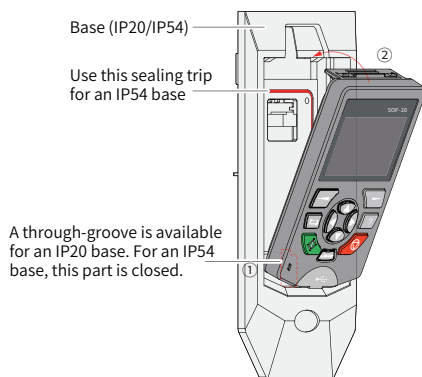


Figure 3-3 Installing SOP-20-880 to the base

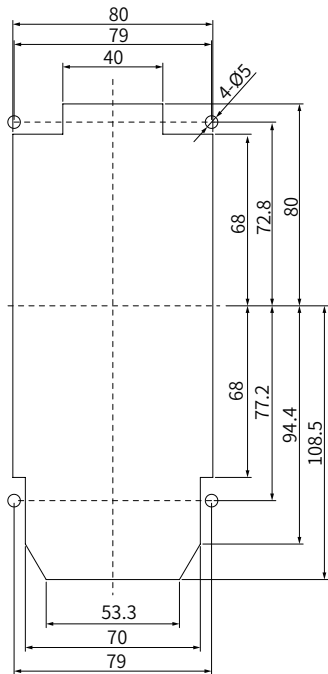


Figure 3-4 Sheet metal slot dimensions
(unit: mm)

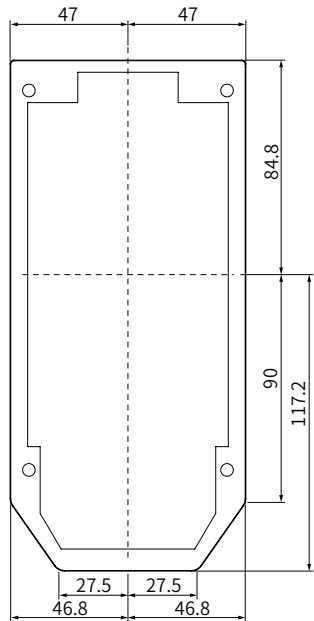


Figure 3-5 Mounting hole dimensions (in mm)

3.3 Connecting SOP-20-880 to the Device

The RJ45 terminal is on the back of SOP-20-880, which is connected to the drive control board through a network cable that compliant with EIA/TIA/568A or 568B standards. Select the termination resistor on the control board with the jumper. It is recommended to use shielded twisted pair cable as SOP-20-880 bus. The maximum length in total is 100 m.

SOP-20-880 networking example:

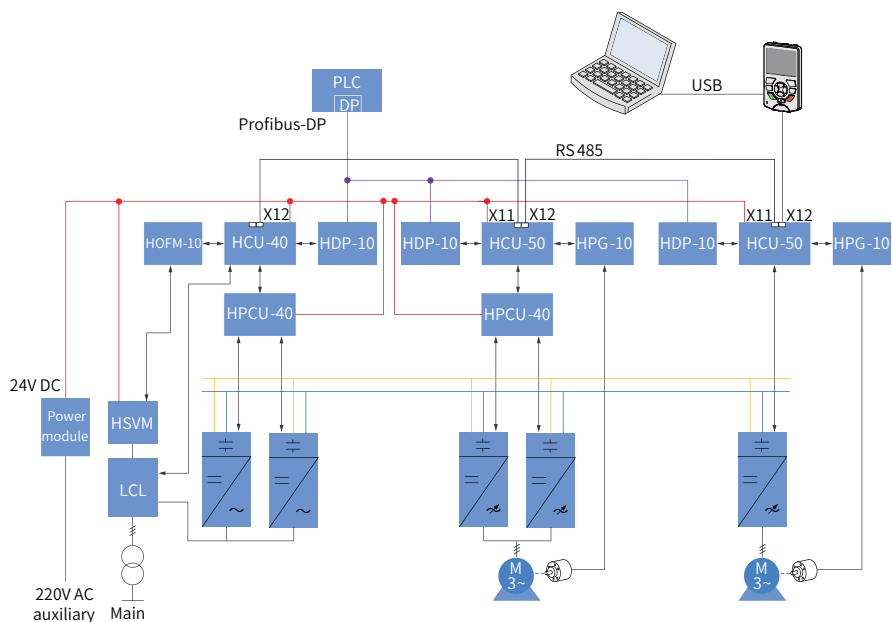


Figure 3-6 Networking example



NOTE

External operating panel is not supported when IP54 installation kit is used.

3.4 Connecting SOP-20-880 to PC

The mini-B type USB connector of SOP-20-880 is used to connect PC. The USB cable length cannot exceed 3 m. The USB2.0 communication protocol is used between SOP-20-880 and PC.

- ① Open the cover of the USB terminal.

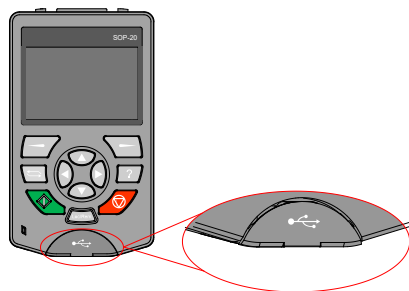


Figure 3-7 Connecting SOP-20-880 to PC (step 1)

- ② Plug in the USB cable and connect the cable to the PC. It is recommended to use cables with magnetic rings.

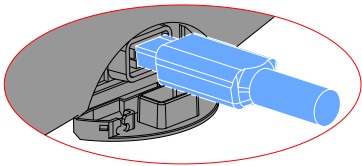


Figure 3-8 Connecting SOP-20-880 to PC (step 2)



NOTE

USB relay mode and USB disk mode are available when the USB cable is connected to PC.

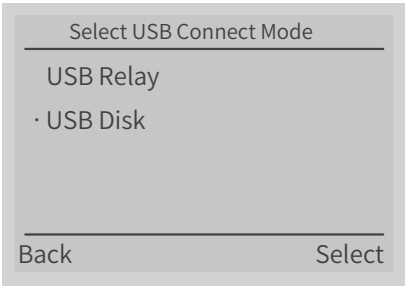


Figure 3-9 Display of SOP-20-880 after being connected to PC

In USB relay mode, SOP-20-880 is used for interaction between the PC tool and the drive.

After selecting the USB disk mode, SOP-20-880 can be used as a memory.

4 Functions and Applications

4.1 Overview

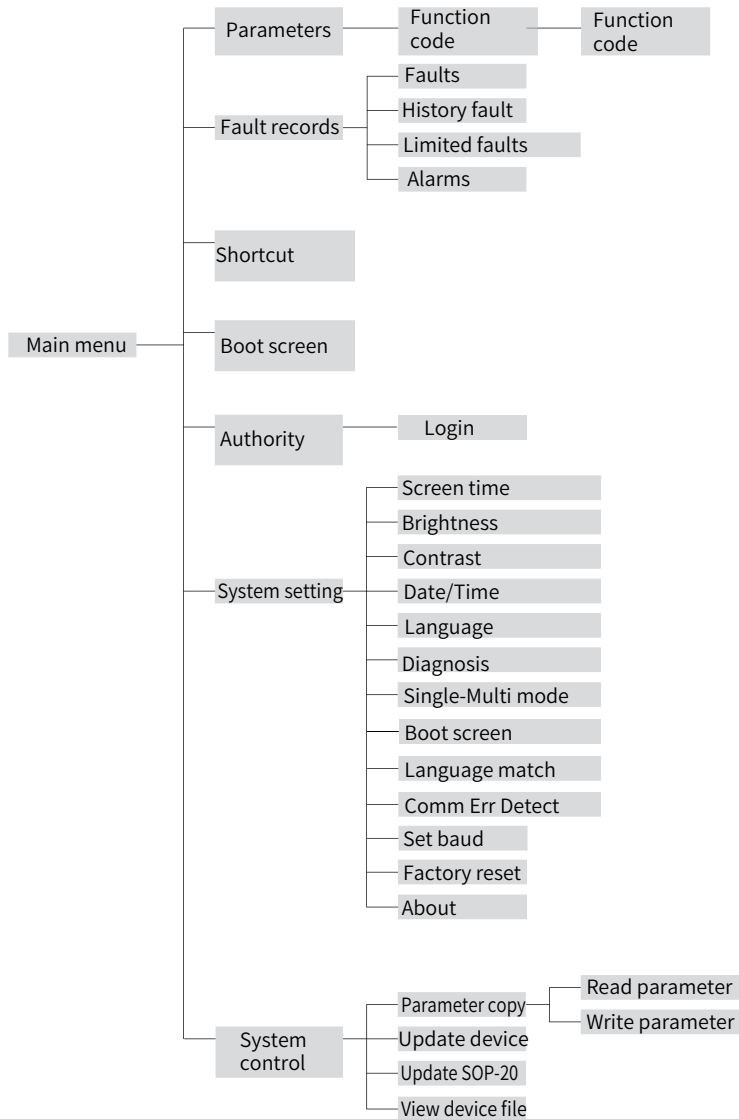


Figure 4-1 Menu structure

4.1.1 Homepage

Homepage: Used to monitor signals. The interface is as below:

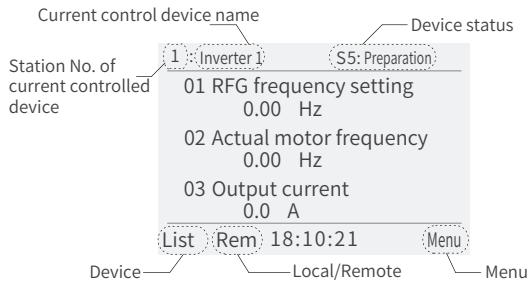


Figure 4-2 Main Interface

- Name of controlled device: Displays the name of the device currently controlled by SOP-20-880.
- Station No. of controlled device: Displays the station number of the device currently controlled by SOP-20-880.
- Device status: Displays the status of the current device, including state machine, faults, and alarms.
- Local/Remote: Displays the current control permission of SOP-20-880. When "Loc" is displayed, SOP-20-880 is granted with the control permission.
- Device: Used to enter the device list in the network where SOP-20-880 is connected to.
- Menu: Used to access the function menu options of SOP-20-880.

You can monitor the device state through the homepage, including the target frequency, actual frequency, output current. Up to three monitored values can be displayed in one page.

1	Inverter 1	S5: Preparation
01 RFG frequency setting		
0.00 Hz		
02 Actual motor frequency		
0.00 Hz		
03 Output current		
0.0 A		
List	Rem	18:10:21
Menu		

Figure 4-3 Display of homepage

The homepage is displayed by default upon power-on. If no operation is performed in interfaces except the homepage within 15 min, the operating panel returns to the homepage automatically.

4.1.2 Start/Stop Control of the Device

Table 4-1 Start/Stop control

Function	Key	Description
Switching between local control and remote control		Press the "Loc/Rem" key to switch between local control and remote control. The bottom left of the interface displays the current control permission. When "Loc" is displayed, SOP-20-880 is granted with the control permission, allowing you to start/stop the device through SOP-20-880. The control permission cannot be switched during operation of the device.
Starting device		Press this key to start the device in the local control mode. This key is invalid in the remote control mode.
Stopping device		Press this key to stop the device in the local control mode. This key is invalid in the remote control mode.
Speed up		Press this key to increase the target speed in the local control mode. This key is invalid in the remote control mode.
Speed down		Press this key to decrease the target speed in the local control mode. This key is invalid in the remote control mode.

4.1.3 Faults

Table 4-2 Fault description

Function	Key	Description
Viewing current fault	Multiple keys	Go to "Menu" > "Fault records" > "Faults".
Viewing current minor faults	Multiple keys	Go to "Menu" > "Fault records" > "Limited Faults".
Viewing current alarms	Multiple keys	Go to "Menu" > "Fault records" > "Alarms".
Viewing the help information of the fault		Select a fault and press the help key to display the help information about the fault.
Fault reset		This function is not related to the control permission. Pressing this key once triggers fault reset once.

Viewing previous faults	Multiple keys	Go to "Menu" > "Fault records" > "History Faults".
-------------------------	---------------	--

4.1.4 Switching the Device

Press  or  on the homepage to enter the online device list interface.

The device list interface lists the devices and their status information in the current system. Use  and  to select the device to be operated, and press  to wait until SOP-20-880 is done operating.

When the device has been switched, the upper left corner of the interface displays specific information about the current device.

4.1.5 Connecting to the PC

SOP-20-880 is connected to the PC through an USB cable either in USB relay mode or in USB disk mode. Do not operate on SOP-20-880 when connecting it to the PC.

Steps for connecting to PC:

- ① Insert one end of the USB cable to the USB port of SOP-20-880, and the other end to the USB port of the PC.
- ② Select the connection mode needed in the connection mode interface.
- USB relay mode: Used for USB-to-485 conversion and supports online commissioning of the device through InoDriveStudio.
- USB disk mode: SOP-20-880 can be used as a removable disk. You can read/write files in SOP-20-880 through the PC.

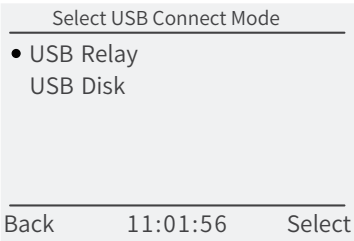


Figure 4-4 Selecting the mode for connecting to PC

4.2 Parameter Setting

Table 4-3 Parameter setting

Function	Key	Description
Viewing or modifying parameters	Multiple keys	Go to "Menu" > "Parameters" → Select the parameter group to be modified → Select the parameter to be modified → View or modify the parameter → Click "Back" or "OK".
Adding or deleting shortcut parameters	 	Used to add a parameter to the shortcut interface In the parameter setting interface, enter any parameter group, move the cursor to the specific parameter, and press the left and help keys at the same time for more than 1.5 seconds to add the parameter to the shortcut interface. Used to delete a parameter from the shortcut interface In the shortcut interface, move the cursor to the parameter to be deleted, press  and  at the same time for more than 1.5 seconds to delete the parameter from the shortcut interface.
Clearing parameters from the shortcut interface	 	Used to clear parameters from the shortcut interface In the quick commissioning interface, press  and  at the same time for more than 1.5s to clear all the parameters in the quick commissioning interface.
Viewing quick commissioning parameters	Go to "Menu" → "Shortcut".	-

4.2.1 Viewing or Modifying Parameters

You can view and modify parameters in the parameter setting interface. The displayed parameters will be different under different control permissions.

If a parameter group is selected, all the parameters under this group will be displayed. You can view or edit these parameters as needed. If the cursor points to a parameter, the following message is displayed. Press  to display the help information.

For read-only parameters, the "Select" button will not be displayed at the right bottom on the screen and  is invalid.

If "0x" is displayed before the parameter, the parameter is a hexadecimal value.

1. Modifying parameters that used to set values

For example, to set C7-04 (namely, ramp 1 acceleration time) to 30.00s, do as follows:

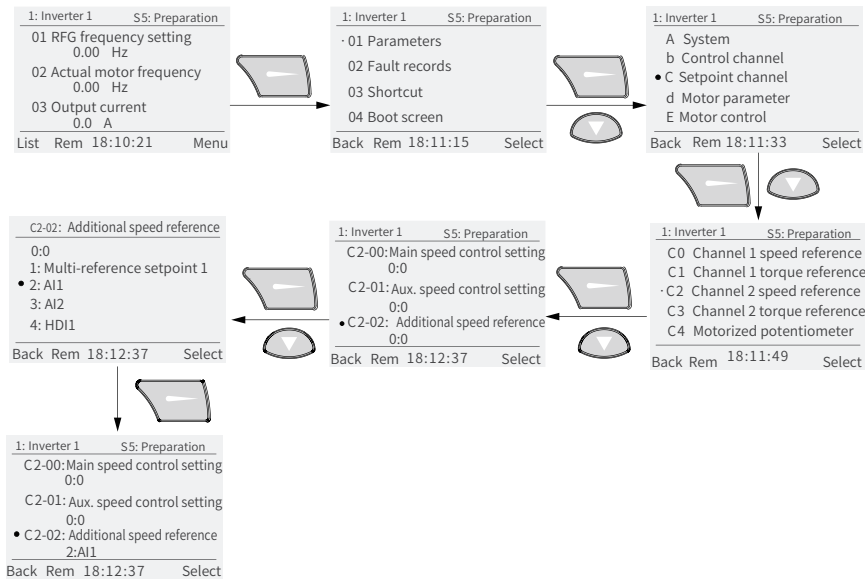


Figure 4-5 Modifying parameters that used to set values

After setting is done, save the setpoint and exit by pressing  or abort the setpoint and exit by pressing .

Press  and  at the same time for more than 1.5 seconds to restore the parameter to default value and then press  to activate the default value.

2. Modifying parameters that used for selection

For example, to set C0-02 (Additional speed reference source) to 2 (A11), do as follows:

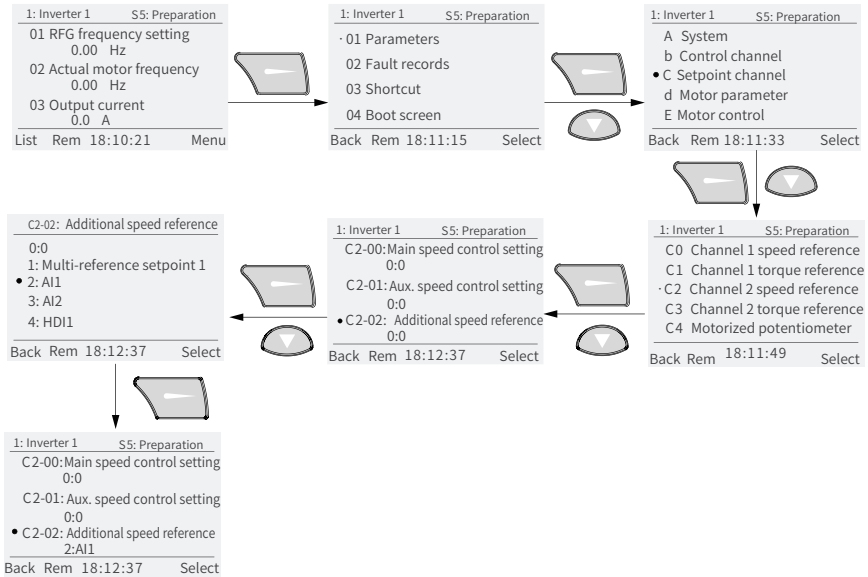


Figure 4-6 Modifying parameters that used for selection

After setting is done, save the setpoint and exit by pressing  , or abort the setpoint and exit by pressing .

Press  and  at the same time for more than 1.5 seconds to restore the parameter to default value and then press  to activate the default value.

4.3 Fault Records

You can view current faults, previous faults, current minor faults, and current alarms in the fault status menu. See section 4.1.3 for details.

4.4 Shortcut

1. Adding shortcut parameters

In the parameter list interface, move the cursor to the parameter to be added and hold down  and  for 1.5s to add the parameter to the shortcut interface. you can add up to 30 parameters to the shortcut interface and a prompt will appear to indicate whether the parameter is added successfully or not.

The parameters in the shortcut interface are in chronological order based on the time when they are added.

To add A0-00 to the shortcut interface, do as follows:

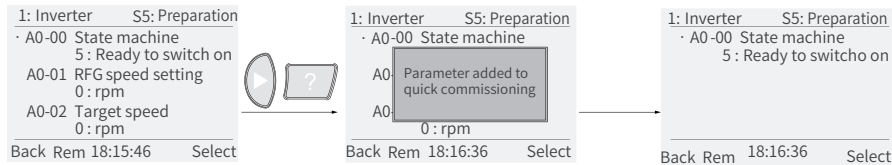


Figure 4-7 Adding shortcut parameters

2. Deleting shortcut parameters

In the shortcut interface, move the cursor to the parameter to be deleted, press and at the same time to delete the parameter from the shortcut interface.

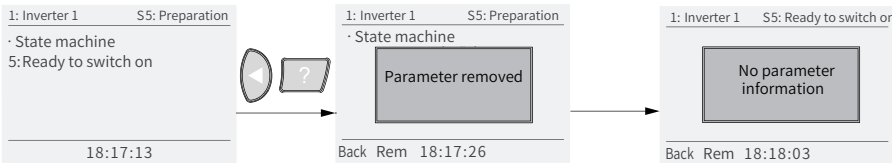
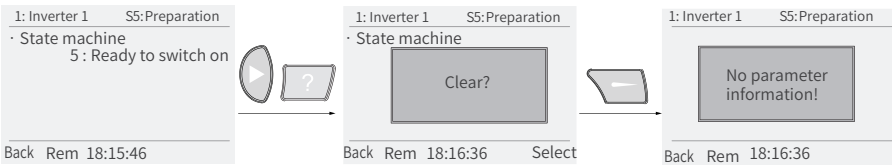


Figure 4-8 Deleting shortcut parameters

3. Clearing shortcut parameters

In the shortcut interface, press and at the same time for more than 1.5 seconds to clear all the parameters from the shortcut interface.



4.4.1 Boot Screen

1. Adding parameters for boot screen

In the parameter list interface, press and for more than 1.5s to add the parameter to the boot screen. You can add up to 3 parameters to the interface and a prompt will be displayed to indicate whether the parameter is added successfully.

Parameters in the boot screen are in chronological order based on the time when they are added.

For example, to add A0-00 to the boot screen, do as follows.

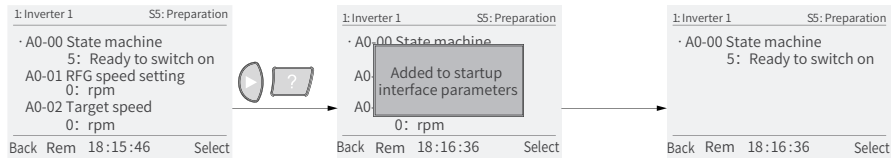


Figure 4-9 Adding boot screen parameters

2. Deleting boot screen parameters

In the boot screen, move the cursor to the parameter to be deleted, and press and for 1.5s to delete the parameter.

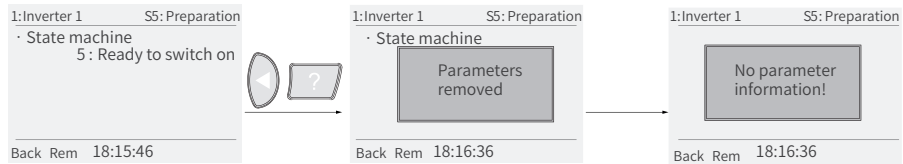


Figure 4-10 Deleting boot screen parameters

3. Clearing boot screen parameters

In the boot screen, press and for more than 1.5 seconds to clear all the parameters from the boot screen.

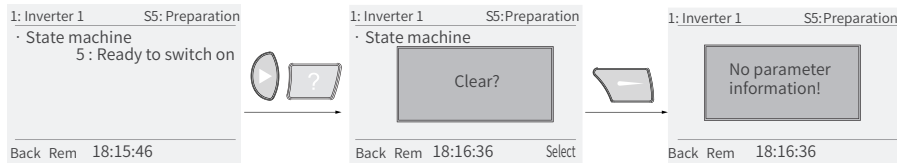


Figure 4-11 Clearing boot screen parameters

4.5 Authority

4.5.1 System Login

Table 4-4 Login description

Function	Description
Logging into the system	Open "Menu" → "Authority" → "Login".

You have to log in to SOP-20-880 before using the following functions under system control:

- Update Device: Used to update the device programs through SOP-20-880.

■ SOP-20-880 upgrade: SOP-20-880 upgrades its programs.

Input a correct password to log in SOP-20-880.

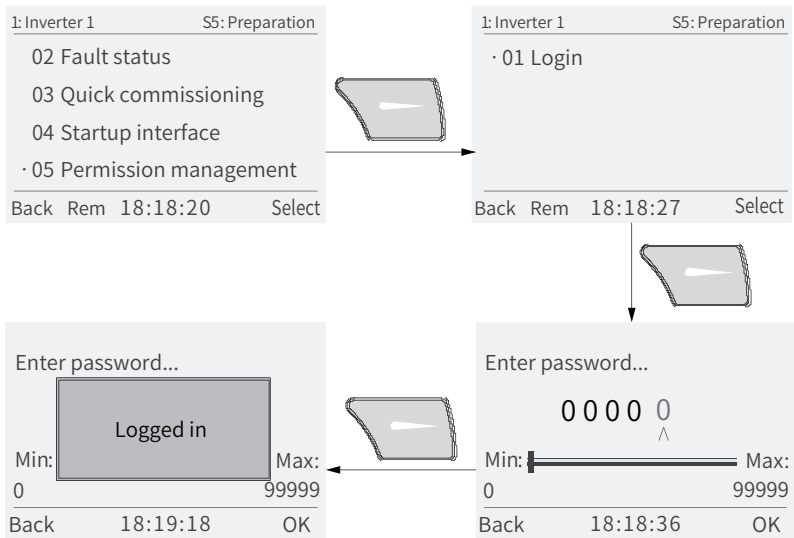


Figure 4-9 SOP-20-880 login

4.6 System Setting

In the system setting interface, you can set the screen saver time, system clock, and language.

Table 4-5 Description of system setting

Function	Description
Setting the screen saver time	Open "Menu" → "System Setting" → "Screen Time".
Adjusting the backlight brightness	Open "Menu" → "System Setting" → "Brightness".
Adjusting the contrast ratio	Open "Menu" → "System Setting" → "Contrast".
Change time	Open "Menu" → "System Setting" → "Date/Time".
Changing the language	Open "Menu" → "System Setting" → "Language".
Network diagnosis	Open "Menu" → "System Setting" → "Diagnosis".
Single-drive/Multi-drive	Open "Menu" → "System Setting" → "Single-Multi Mode".
Boot screen	Open "Menu" → "System Setting", and "Boot screen".
Language matching check	Open "Menu" → "System Setting" → "Language match".
Connection error detection	Open "Menu" → "System Setting" → "Comm Err Detect".
Baud rate setting	Open "Menu" → "System Setting" → "Set Baud".

Restoring default values	Open "Menu" → "System Setting" → "Factory Reset".
System information	Open "Menu" → "System Setting" → "About".

1. Setting screen saver time

The backlight can be turned off when no operation is performed on keys for a period of time. The time unit is minute. You can set to 99 min at maximum. When it is set to 0, the screen saver function is turned off. When the backlight is turned off automatically, pressing any key wakes up the backlight again.



NOTE

After the backlight is turned off automatically, if a new fault occurs, the backlight turns on again. When the set screen saver time elapses, the backlight turns off again.

2. Adjusting the backlight

Adjust the brightness of the LCD backlight. The higher the value, the higher the backlight is.

3. Contrast ratio

Adjust the contrast between the words and the background. The higher the value, the stronger the difference between the words and the background.

4. Language

Supports switchover between Chinese and English.

5. Date/Time

① SOP-20-880 time

Used to modify the date and time of SOP-20-880.

② Device object time

Get the time of the currently connected device and modify the time.

③ Synchronous device time

Write the current date and time of SOP-20-880 to the currently connected device to synchronize the device with the time of SOP-20-880.

6. Network diagnosis

Used to detect whether the baud rates of multiple device objects are consistent. If not, a window will pop up indicating the baud rate of each device.



NOTE

The station numbers of different devices in the same network must be different. Network diagnosis cannot be performed when station numbers are the same.

7. Single-drive/Multi-drive

The switchover between single-drive and multi-drive modes is supported.

- ④ Single-drive mode: When the SOP-20-880 is powered on for scanning, the scan stops immediately after any device responds. The SOP-20-880 only interacts with this device.
- ⑤ Multi-drive mode: When the SOP-20-880 is powered on for scanning, stations 1 to 127 will be scanned. SOP-20-880 can be connected to any device that interacted successfully during the scan cycle. (Note: Only one device can be interacted with at the same moment.)

8. Boot screen

- ① Displays the manufacturer logo: When the system is powered on, the Inovance logo is displayed.

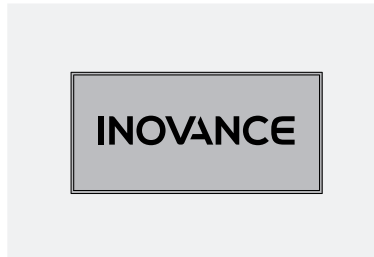


Figure 4-10 Manufacturer logo displayed

- ② Manufacturer logo not displayed: When the system is powered on, the text "SOP-20" is displayed.

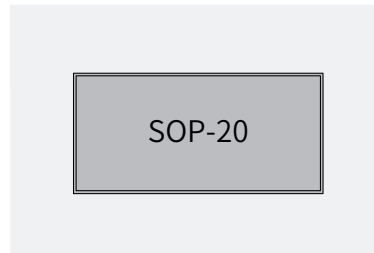


Figure 4-11 Manufacturer logo not displayed

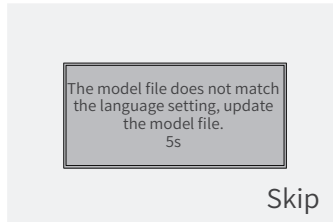
9. Language matching check

The setting is performed by the user through "Menu" → "System Setting" → "Language Match".

This parameter is saved after power-off (enabled by default).

If you set this parameter to "Disable", no check will be performed. If you set this parameter to "Enable" and the valid model file does not match the language set by the operating panel, the device scan will be completed. Meanwhile, the message "Update the model file if the model file does not match the operating panel language" is displayed, as shown in the following figure. If the file language of the matching model

is the same as that of SOP-20-880, no pop-up window will be displayed. The window disappears automatically after 8s. You can also click the  to close the pop-up window.



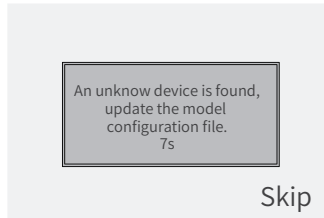
10. Connection error detection

You can set this through "Menu" → "System setting" → "Comm Err Detect".

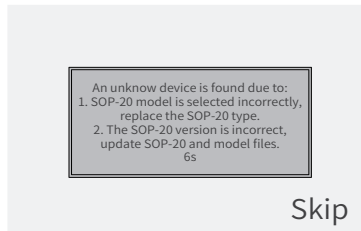


This parameter is saved after power-off (enabled by default).
If you set this parameter to "Disable", no detection will be performed.

- ① If you set this parameter to "Enable", and SOP-20-880 lacks one or more device model files in the cascaded network, resulting in device connection error, the pop-up window will be displayed as shown in the following figure.



- ② If you set this parameter to "Enable", SOP-20-880 does not support one or more devices in the cascaded network, resulting in device connection error, the pop-up window will be displayed as shown in the following figure.



The window disappears automatically after 8s. You can also click the  to close the window.



NOTE

If there is no countdown or "Skip" in the connection error window, no device in the cascaded network can interact with SOP-20-880.

11. Baud rate setting

This option can be used to modify the baud rate of all devices in the cascaded network.

12. Parameter initialization

After confirming the operation, you can restore all parameters in the "System setting" to default values.

13. System information

Used to view the version information of the SOP-20-880.

4.7 System Control

You have to log in SOP-20-880 before using functions such as parameter copy and device update. For details of login method, see section 4.5.1.

Table 4-6 System operation

Function	Description
Parameter copy	Open "Menu" → "System Control" → "Parameter Copy", and select "Read Parameter" or "Write Parameter".
Programming	Open "Menu" → "System Control" → "Update Device". Make sure you have already logged in the system before this operation.
Upgrading SOP-20	Open "Menu" → "System Control" → "Update SOP-20". Make sure you have already logged in the system before this operation.
Viewing the model file	Open "Menu" → "System Control" → "View Device File".

4.7.1 Parameter Copy

You can use the parameter copy function to transfer the parameter settings of one device to the other device. Upload and save the device parameters that have done commissioning, and then download them to the target device. The device that reads/writes parameters is the device being operated by SOP-20-880 currently, as shown in the upper left corner. After login, access the parameter copy interface.

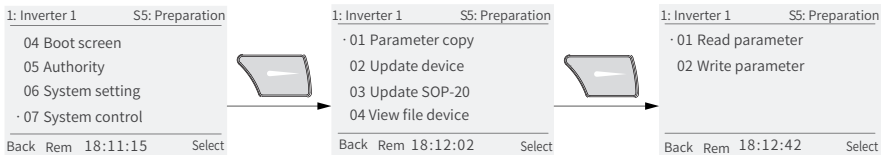


Figure 4-10 Parameter copy

1) Reading parameters

Set the user-defined part of the file name before reading parameters. The user-defined part consists of 5 digits. Use the left and right direction keys to select the digit to be set, and up and down direction keys to set character values. The range of the characters is 0 to 9 and A to Z. The format of the file name is as follows: Device name - 5-digit user-defined name

Example: Inverter-12ABC.ipv

Note: The device is an inverter and the customized field name is 12ABC.

Press  to read and copy the parameters after the file name is set, the message "Parameters are copied successfully" will be displayed on the interface.

Specific reasons will be displayed if parameter-read/write fails.

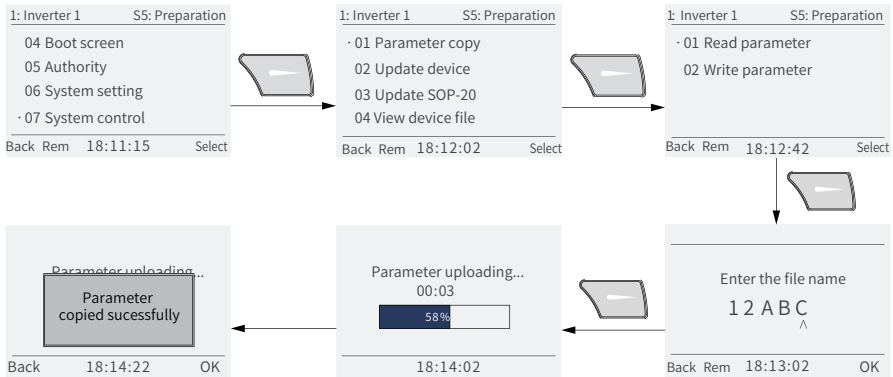


Figure 4-11 Example of reading parameters

2) Writing parameters

The procedure for writing parameters is as below:

- ① Press  to select the target device.
- ② Enter "Write Parameter" menu, then press  to select the parameter file that has been read.
- ③ In the copy mode interface, select the parameter copy mode you need.
- ④ Wait until the parameter download is done.

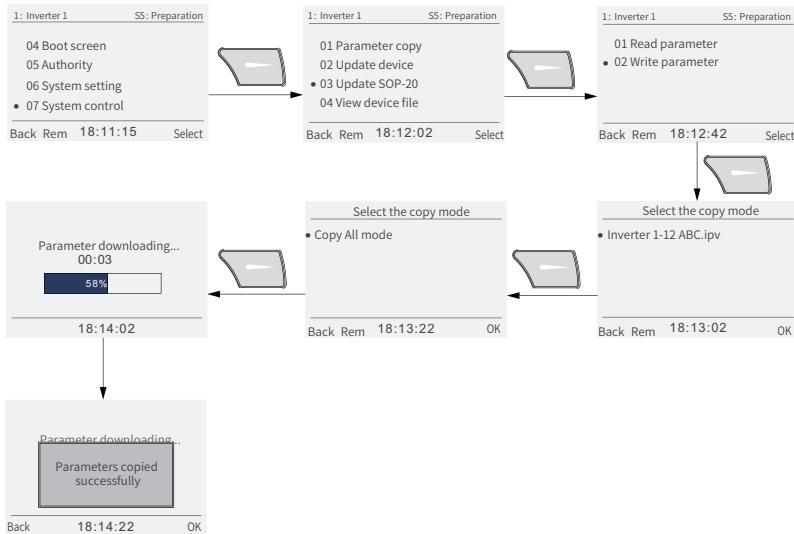


Figure 4-12 Example of writing parameters



All parameter files that copied, saved and read are stored in SD card. Ensure that the SD card is properly installed before using the parameter copy function.

Example: Copy all the parameters in "Inverter 1" to "Inverter 2".

- ① Press to enter the device list and select "Inverter 1".
- ② Go to "Authority", enter the correct password, and log in to SOP-20-880.
- ③ Enter "System Control" to select "Parameter Copy".
- ④ Select "Read parameter" and name it "12ABC".
- ⑤ After confirmation, wait until parameters are done reading.
- ⑥ When "Inverter 2" is connected to the SOP-20-880 network, press to enter the device list. Select "Inverter 2" (or connect SOP-20-880 directly to "Inverter 2").
- ⑦ Open "System Control" → "Parameter Copy" → "Write parameter".
- ⑧ Select the parameter file to be written, and then select the copy mode.
- ⑨ Wait until parameters are downloaded.

4.7.2 Updating the Device

You may upgrade the program directly through SOP-20-880. The following shows the software upgrade procedure.

- ① Prepare the device upgrade file.
- For example, when SOP-20-880 is connected to the PC, select "USB Disk".

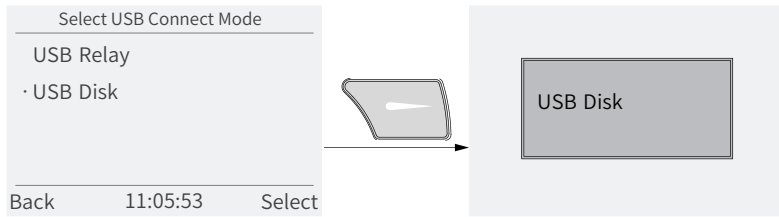


Figure 4-13 USB connected

- Copy the upgrade file to "X:\SYS\PROGRAM" of SOP-20-880.
 - ② Disconnect SOP-20-880 from the PC and connect SOP-20-880 to the device using an RJ45 cable.
 - ③ Enter "Login" in "Authority" and enter the correct password to obtain the upgrade permission.
 - ④ Enter "Update Device" in "System Control".
 - ⑤ Select the file to be upgraded and press  after confirming the file information is correct.
 - ⑥ Set the station number for the programming device.
 - ⑦ Select the baud rate (115200 bps by default) for programming.
 - ⑧ Select whether to start operation when the programming is done.
 - ⑨ Wait until programming is done.

4.7.3 Upgrading SOP-20-880

The software can be upgraded through SOP-20-880. The following shows the software upgrade.

- ① Prepare the SOP-20-880 upgrade file.
- For example, when SOP-20-880 is connected to the PC, select "USB Disk".

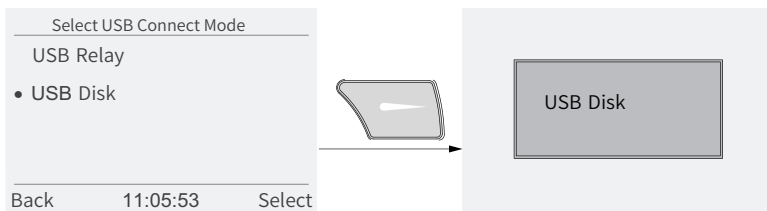


Figure 4-14 USB connected

- Copy the upgrade file to "X:\SYS\Update" of SOP-20-880.
 - ② Disconnect SOP-20-880 from the PC and connect SOP-20-880 to the device (power supply) through the RJ45 cable.
 - ③ Enter "Authority" → "Login", enter the correct password and obtain SOP-20-880 upgrade permission.

- ④ Enter "Update SOP-20" in "System Control".

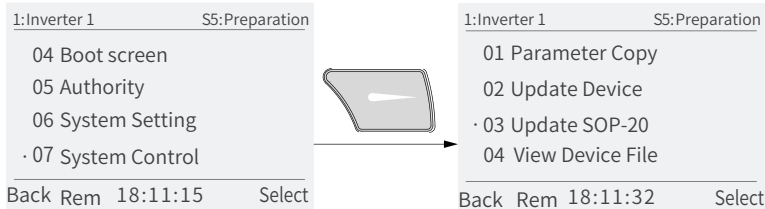


Figure 4-15 Setting the upgrade menu

- ⑤ Select the files to be upgraded.

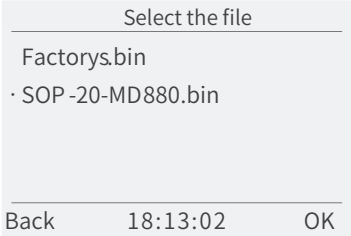


Figure 4-16 Select the files to be upgraded.

- ⑥ Press  in the pop-up window.

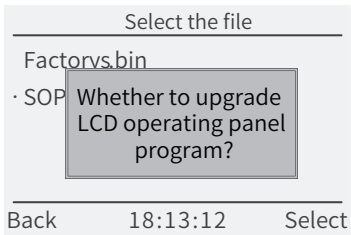


Figure 4-17 Confirming upgrade

- ⑦ Wait until SOP-20-880 restarts.

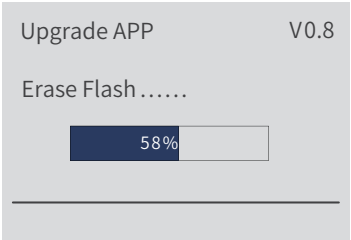


Figure 4-18 Upgrade in progress

4.7.4 Viewing Model Files

You can view all the valid model file names, RFIDs, and languages of SOP-20-880.

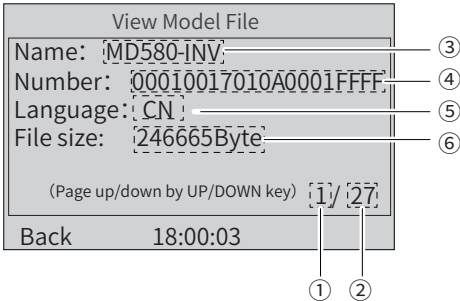


Figure 4-12 Example of model file interface

Table 4-7 Interface description

No.	Display	Description
①	No.	Current model file No.
②	Sum	Number of valid model files
③	Name	Model file name
④	Number	Device E-tag corresponding to the model file
⑤	Language	Parameter list language corresponding to the model file
⑥	File size	Current size of the model file

4.8 Multi-drive Application

When a system consisting of multiple devices communicates with SOP-20-880, the station numbers of different devices cannot be the same, and the communication baud rate must be the same. For example, all devices must be set to 4 Mbps. The default station numbers of the devices are the same. For the first time, you need to set the corresponding station numbers through SOP-20-880.

- 1. Setting the device station number and communication baud rate:
 - ① Disconnect the device that needs to be assigned with the station number from other devices and connect SOP-20-880 separately.
 - ② After SOP-20-880 searches for the device, change the station number of the device through A10-02 and set the communication baud rate through A10-03.
- 2. Connecting multiple devices to SOP-20-880:
 - ① Connect RJ45 interface of SOP-20-880 to the first device through a network cable that meets the standards of EIA/TIA/568A or 568B.
 - ② Use daisy-chain wiring mode to connect the second interface of the first device to

the second device, connect the second device to the third device, and so on. The final device matches the termination resistor through a jumper.

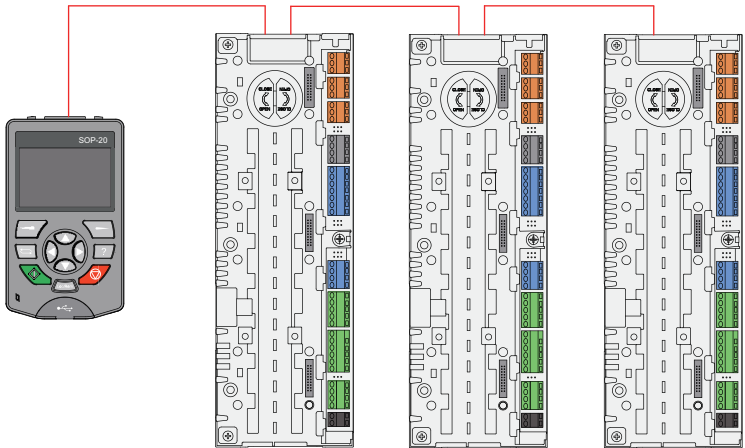


Figure 4-19 Connecting SOP-20-880 to multiple devices

- ③ Connect SOP-20-880 and wait for SOP-20-880 to search for devices automatically.
3. Viewing status of multiple devices:

Press  to enter the device status display page. You can view the device name and current status.

1: Inverter 1		S5: Preparation	
· 1: Inverter 1		Stop	
2: Inverter 2		Stop	
Back		Rem 18:11:15	
		Select	



When a device is selected, SOP-20-880 calls the device. The RUN and FAULT indicators of the HCU flash simultaneously.

4. Switching the operating device

Press  or the left soft key to enter the list of online equipment interface. The device list interface lists the devices and their status information in the current system.

After selecting the device to be operated, press the right soft key to wait for SOP-20-880 to perform switchover. When switchover is done, the upper left corner on the page

displays specific information about the current device and station number.

5. Viewing the fault state of the device

In the fault state interface, you can view all faults of current device.

5 Maintenance and Inspection

5.1 Replacing the Battery

- ④ Use a tool or fingers to release the battery rear cover.

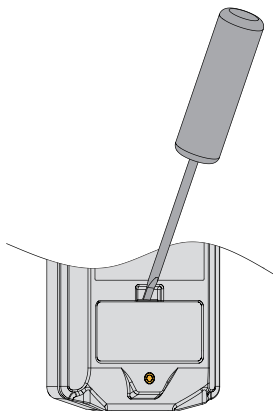


Figure 5-1 Opening the battery rear cover

- ⑤ Take out the battery with tweezers or a slotted screwdriver.

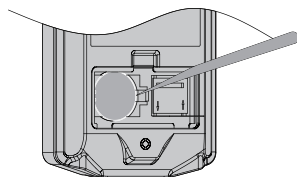


Figure 5-2 Taking out the battery

- ⑥ When installing the battery, insert the battery into the snap-fit joint on end A first, then press end B down to put the battery in place.

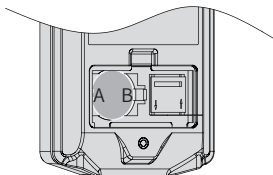


Figure 5-3 Closing the battery cover

- ⑦ Close the battery cover finally.

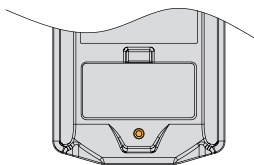


Figure 5-4 Replacing the battery

5.2 Replacing the SD Card

- ① Use a tool or fingers to release the battery rear cover.

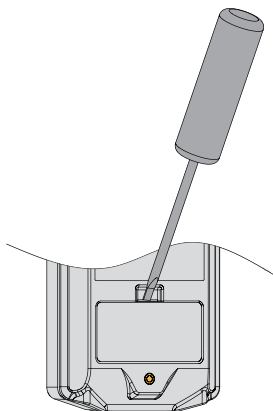


Figure 5-5 Opening the battery rear cover

- ② Push the metal cover towards the unlock direction as shown by the arrow to release the SD card.

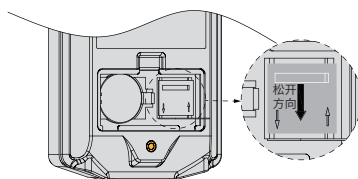


Figure 5-6 Unlock the card holder and remove the SD card.

- ③ Lift up the metal cover and place a new SD card.

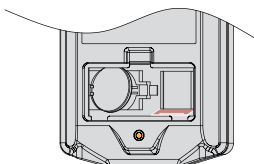


Figure 5-7 Insert the SD card

- ④ Put down the metal cover and push the metal cover towards the lock direction as shown by the arrow to fix the SD card.

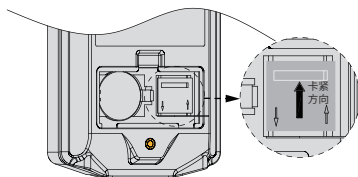


Figure 5-8 Securing the SD card

- ⑤ Put the battery cover back.

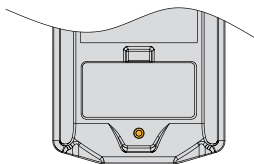


Figure 5-9 SD card has been replaced.



NOTE

Hot-swap is not supported by the SD card. Replace the SD card when SOP-20-880 is powered off.



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